

Options: Synthetic Arbitrage

Conversion & reverse conversion (reversal)

Concept

Every option position has a *synthetic* equivalent built from the other option and the underlying. A **conversion** pairs a real long stock with a synthetic short stock; a **reversal** (reverse conversion) pairs a real short stock with a synthetic long stock. Both create a delta-neutral, locked box whose value depends only on interest and dividends—so a mispriced put/call pair versus spot is harvested directly. These are the executable form of put–call parity.

Synthetic Building Blocks

At common strike K , expiry T :

$$\text{Synthetic long stock} = +C - P, \quad \text{Synthetic short stock} = -C + P.$$

Adding the real underlying gives the two riskless structures below.

Conversion

Position: long stock, short call (K), long put (K) = long stock + synthetic short.

Leg	CF $t=0$	$S_T < K$	$S_T \geq K$
Long stock	$-S_0$	S_T	S_T
Short call	$+C$	0	$-(S_T - K)$
Long put	$-P$	$K - S_T$	0
Net	$C - P - S_0$	K	K

Terminal value is always K . The position is a synthetic bond. Fair pricing requires

$$S_0 - C + P = Ke^{-rT} \iff C - P = S_0 - Ke^{-rT},$$

i.e. put–call parity. **Conversion profit** (per share, PV):

$$\pi_{\text{conv}} = (C - P) - (S_0 - Ke^{-rT})$$

Profitable when $C - P$ is *too high* (call rich / put cheap): $\pi_{\text{conv}} > 0$.

Reversal (Reverse Conversion)

Position: short stock, long call (K), short put (K) = short stock + synthetic long.

Leg	CF $t=0$	$S_T < K$	$S_T \geq K$
Short stock	$+S_0$	$-S_T$	$-S_T$
Long call	$-C$	0	$S_T - K$
Short put	$+P$	$-(K - S_T)$	0
Net	$S_0 - C + P$	$-K$	$-K$

You pay a certain K at expiry, collecting $S_0 - C + P$ now. **Reversal profit** (PV):

$$\pi_{\text{rev}} = (S_0 - Ke^{-rT}) - (C - P) = -\pi_{\text{conv}}$$

Profitable when $C - P$ is *too low* (put rich / call cheap).

With Dividends

If the underlying pays present-value dividends D before expiry, replace S_0 by $S_0 - D$ in the fair relation:

$$\pi_{\text{conv}} = (C - P) - (S_0 - D - Ke^{-rT}).$$

Dividends received on the long-stock leg of a conversion improve it; borrow costs on the short-stock leg of a reversal worsen it.

Choosing the Trade

Signal	Trade	Direction
$C - P > S_0 - Ke^{-rT}$	Conversion	long stock, short call, long put
$C - P < S_0 - Ke^{-rT}$	Reversal	short stock, long call, short put

Worked Example

$S_0 = 50$, $K = 50$, $r = 4\%$, $T = 0.25$, no dividends. Fair $S_0 - Ke^{-rT} = 50 - 50e^{-0.01} = 0.4975$.
Market: $C = 2.10$, $P = 1.40$, so $C - P = 0.70$.

$$\pi_{\text{conv}} = 0.70 - 0.4975 = 0.2025 \text{ per share } (\$20.25/\text{contract}).$$

Since $C - P$ is too high, run the **conversion**: long stock, short call, long put.

Practical Notes

- Delta-neutral by construction; residual risk is early assignment (American options) and pin risk at K .
- Financing (short-stock rebate, margin interest) and dividends dominate the tiny edge— model them exactly.
- Execute all three legs together (a “conversion/reversal” combo order) to avoid legging risk.